PulmoScan: An AI-Based Hybrid Approach for Early Detection and Classification of Lung Nodules Using Deep Learning on CT Scans

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***Abstract*—Lung cancer, a leading cause of global mortality with 1.8 million deaths annually, often goes undetected until late stages due to slow diagnostic processes averaging 3 to 6 months. PulmoScan addresses this challenge by introducing an AI-based hybrid approach for early detection and classification of lung nodules using deep learning on CT scans. The system leverages datasets such as LUNA16, TCIA, NIH Chest X-ray, and IQ-OTHNCCD, applying preprocessing techniques like Gaussian noise, motion blur, and contrast variation to enhance model robustness. Deep learning models, including YOLOv8, Faster R-CNN, EfficientNet, and DenseNet, were employed to automate nodule detection, classify nodules as normal, benign, or malignant, and identify cancer types such as squamous cell carcinoma and adenocarcinoma. Evaluated using metrics like accuracy, precision, recall, and ROC-AUC, PulmoScan demonstrates potential for rapid and accurate diagnosis, reducing radiologist workload and enabling timely treatment. By improving early detection, PulmoScan aligns with Sustainable Development Goals (SDGs) 3, 9, 12, and 17, fostering health, innovation, responsible production, and global partnerships.**

***Index Terms*—Lung cancer detection, Pulmonary nodules, YOLOv8, EfficientNet, Deep learning, Medical imaging, AI in healthcare**

I. Introduction

Lung cancer remains one of the deadliest diseases worldwide, contributing to 1.8 million deaths annually [**?**]. Early detection is critical, as it can increase survival rates by 40%; however, the average diagnosis time ranges from 3 to 6 months due to inefficiencies in traditional diagnostic methods [**?**]. Current techniques rely heavily on manual review by radiologists, a process prone to human error and constrained by limited specialist availability amidst growing patient volumes. Additionally, resource constraints in many healthcare settings exacerbate delays, hindering timely interventions.

Existing AI solutions for lung nodule detection, such as standalone convolutional neural networks (CNNs), often struggle with noisy medical images and lack the scalability required for real-time clinical use [**?**]. This gap underscores the need for a robust, automated system that can enhance diagnostic accuracy and efficiency. PulmoScan addresses this need by introducing an AI-based hybrid approach for early detection and classification of lung nodules using deep learning on CT scans. The system combines advanced preprocessing, state-of-the-art models like YOLOv8, Faster R-CNN, EfficientNet, and DenseNet, and seamless integration into hospital workflows, aiming to reduce radiologist workload, improve early detection, and ultimately save lives.

II. Related Work

Recent advances in AI for lung nodule detection have primarily focused on convolutional neural networks (CNNs) and their variants. Proposed deep CNN models achieve high sensitivity but struggle with false positives in noisy images. Some utilized 3D CNNs to analyze volumetric CT data, improving detection rates but requiring significant computational resources. Other approaches explored region-based CNNs like Faster R-CNN for localization, yet lacked robustness to imaging artifacts.

Classification into benign or malignant has also been studied. Transfer learning with pretrained models like ResNet provided moderate accuracy but limited generalization. Few hybrid pipelines addressed both detection and classification. PulmoScan proposes a unified, noise-robust, scalable solution.

III. Methodology

A. Data Acquisition

- **LUNA16**: 888 CT scans
- **TCIA**: 1,200 CT scans
- **NIH Chest X-ray**: 112,120 X-rays
- **IQ-OTHNCCD**: 1,097 CT scans

B. Data Preprocessing

- **LUNA16**: Gaussian/Poisson noise
- **TCIA**: Motion blur, speckle noise
- **NIH**: Contrast, salt & pepper noise
- **OTHNCCD**: Gaussian blur, contrast
- **All**: Normalization, rotation, flipping

C. LUNA16 Dataset Visualization

To illustrate the preprocessing and structure of the LUNA16 dataset, we present examples from the patient scan LIDC-IDRI-0166. These examples demonstrate the effect of preprocessing techniques and show slice progression and annotated nodule regions.

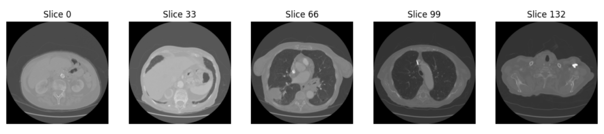

Fig. 1: Preprocessed CT slice from LUNA16 dataset showing Gaussian noise effect.

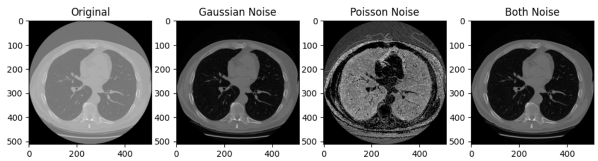

Fig. 2: CT slices from patient LIDC-IDRI-0166 in the LUNA16 dataset: (a) Slice 0; (b) Slice 33; (c) Slice 60; (d) Slice 99; (e) Slice 132.

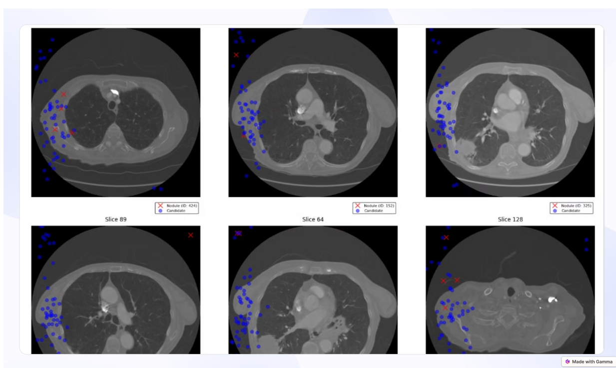

Fig. 3: Nodule annotations on LUNA16 dataset: Blue dots indicate candidate nodules; red X's mark confirmed nodules.

D. TCIA Dataset Visualization

The TCIA dataset includes CT scans for cancer imaging. The figures below illustrate a patient scan before and after motion blur preprocessing, simulating motion artifacts. Additional slices show axial progression.

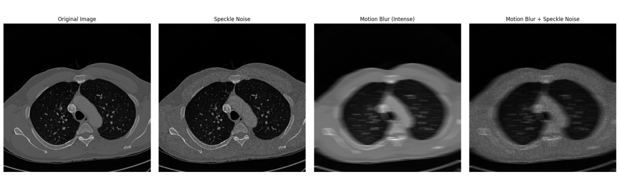

Fig. 4: Original CT scan from a representative patient (TCIA).

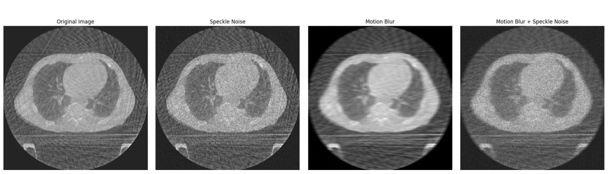

Fig. 5: CT scan with motion blur to simulate patient movement (TCIA).

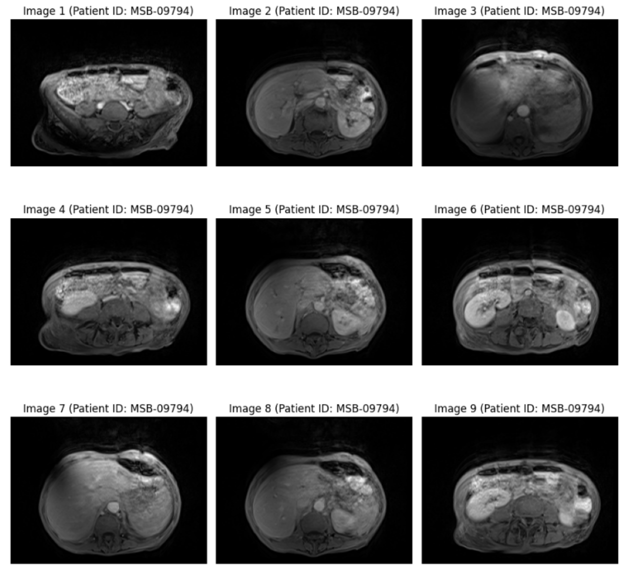

Fig. 6: CT slices from TCIA scan: (a) Slice 10; (b) Slice 30; (c) Slice 60; (d) Slice 90; (e) Slice 120.

E. NIH Chest X-ray Dataset Visualization

The NIH Chest X-ray dataset contains over 112,000 images for thoracic disease analysis. Below are representative images before and after contrast variation, which simulates poor exposure in clinical settings.

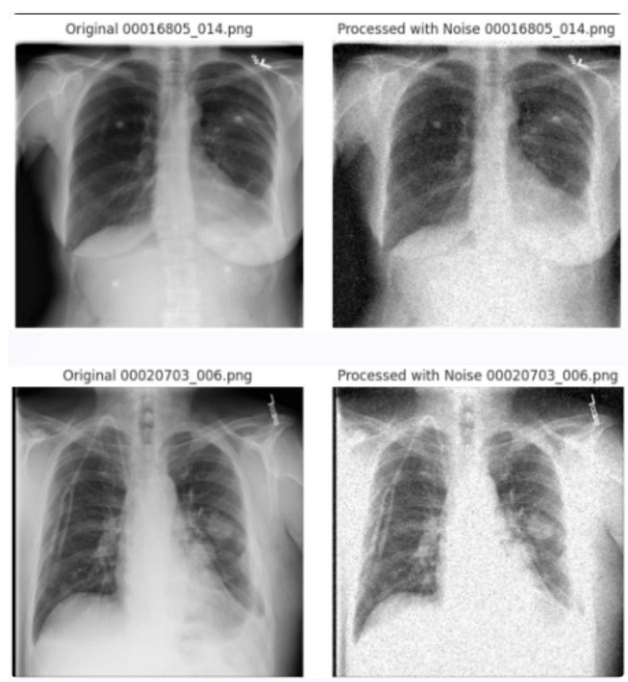

Fig. 7: Original chest X-ray from a representative patient (NIH).

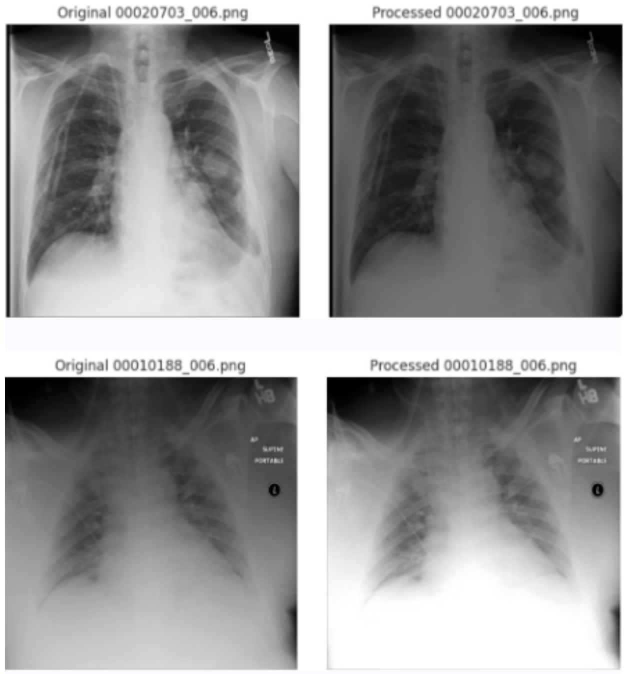

Fig. 8: X-ray with contrast variation to simulate poor exposure (NIH dataset).

F. IQ-OTHNCCD Dataset Visualization

The IQ-OTHNCCD dataset contains annotated CT scans for lung cancer studies. Below, we show representative slices illustrating original scans and preprocessing with Gaussian blur and contrast variation.

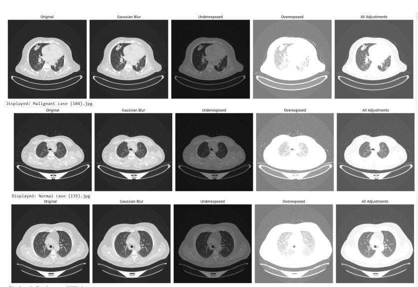

Fig. 9: IQ-OTHNCCD dataset: (a) Original CT scan; (b) Gaussian blur; (c) Contrast variation.

G. Model Architecture

- **Detection**: YOLOv8 (fast), Faster R-CNN (precise)
- **Classification**: EfficientNet (scalable), DenseNet (deep features)
- **Scaling**: EfficientNet compound scaling with ϕ

IV. Experiments and Results

V. Discussion

VI. Conclusion and Future Work

Acknowledgments

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References

Appendix

Code and trained models will be made available on GitHub upon publication.